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Roll No

ME-702(C)-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Power Plant Engineering

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Why thermal plants are not suitable for supplying fluctuating loads? 7
b) What are the different sources of energy available for power generation? How long they can last? 7
2. a) List the four important circuits of the steam power plant. 7
b) Define boiler mountings and accessories. 7
3. a) Draw a general layout of steam power plant with neat diagram and explain the working of different circuits. 7
b) Describe the working of a high pressure boiler with super heaters. 7
4. a) What do you understand by fluidized bed combustion? 7
b) Explain in detail the coal handling system with a suitable block diagram. 7

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5. a) Describe the different types of over feed stokers and discuss its merits and demerits of each over others. 7
b) Why is Coal pulverized? Explain any one type of pulverized systems used now-a-days. 7
6. a) Why is shielding of a Nuclear reactor necessary? 7
b) What are the components of pressurized water reactor nuclear power plant? 7
7. a) List some of the disadvantages of Nuclear power plant. 7
b) What is chain reaction? How it is maintained? What is the difference between controlled and uncontrolled chain reaction? Explain with neat sketches and with examples. 7
8. a) What is meant by load factor and diversity factor? 7
b) Write an explanatory note on the economics of power generation. 7
